

SUMMER TERM 2024
DEPARTMENTALLY MANAGED REMOTE ONLINE EXAMINATION
ECON0039: ADVANCED MACROECONOMICS
Solution:

Assessment Component: 100% Remote Online Controlled Condition Examination

Time allowance

You have 2 hours to complete this examination, plus an additional collation time of 20 minutes and an Upload Window of 20 minutes. The additional collation time has been provided to cover any additional tasks that may be required when collating documents for upload, and the Upload Window is for uploading and correcting any minor mistakes. The additional collation time and Upload Window time allowance should not be used for additional writing time.

If you have been granted SoRA extra time and/or rest breaks, your individual examination duration and additional collation time will be extended pro-rata and you will also have the 20-minute Upload Window added to your individual duration.

If you miss the submission deadline during the 40-minute Late Submission Period and do not receive approved mitigation for the circumstances relating to your late submission, a Late Submission Penalty will be applied by the Module Administrator. At the end of the Late Submission Period, you will not be able to submit your work via Moodle and you will not be permitted to submit the work via email or any other channel.

All work must be submitted anonymously in a single PDF file. Do not write your name and student number in either the file or the file name. The file name must be in the following format: Module Code-%Exam i.e. ECON0039-100%Exam.

Page Limit: No page limit

Academic Misconduct: By submitting this assessment, you are confirming that you have not violated UCL's Assessment Regulations relating to Academic Misconduct contained in [Section 9 of Chapter 6](#) of the Academic Manual.

All questions should be answered.

Part A carries 40% of the total mark, Part B 20% and Part C 40%.

PART A : k/y in a Model with Young and Old

THIS PROBLEM considers a general equilibrium model with production and capital accumulation. Economic agents live for two periods. They are young in the first period of their life and old in the second. In a given period t , there is one young and one old. Population is constant and is equal to two. The utility of an agent born in t is

$$U = \log C_{1,t} + \log C_{2,t+1},$$

where $C_{1,t}$ is the consumption when young (in period t) and $C_{2,t+1}$ the consumption when old ($t+1$). When young, an agent supplies inelastically one unit of labour, and the wage is W_t . Savings are done under the form of capital and when old, agents consume their capital K_{t+1} (capital does not depreciate) plus the return on that capital (rate of return on capital is r_{t+1}). They do not work when old.

There is a representative competitive firm who produces the unique good in the economy (that can be consumed or saved under the form of capital) according to

$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}$$

where Y_t is output (the numéraire), K_t capital, A_t productivity, L_t labour and $0 < \alpha < 1$. Given the previous assumptions, $L_t = 1$ (only the young work), and we assume that productivity grows at rate g :

$$A_{t+1} = (1 + g)A_t$$

A.1 The profit function of the representative firm is $\Pi_t = Y_t - r_t K_t - W_t L_t$. From the first order conditions, derive an expression for W_t and r_t . Rewrite these conditions and the production function in deflated terms (where $k_t = K_t/(A_t L_t)$, $y_t = Y_t/(A_t L_t)$ and $w_t = W_t/(A_t L_t)$).

A.2 The budget constraints of an agent born in t are

$$C_{1,t} = w_t A_t - K_{t+1}$$

in period t and

$$C_{2,t+1} = (1 + r_{t+1})K_{t+1}$$

in period $t+1$. Write down the intertemporal budget constraint of an agent born in t . Find optimal $C_{1,t}$ and $C_{2,t+1}$. Find an expression for the saving rate. Is it a increasing function of the interest rate? Why?

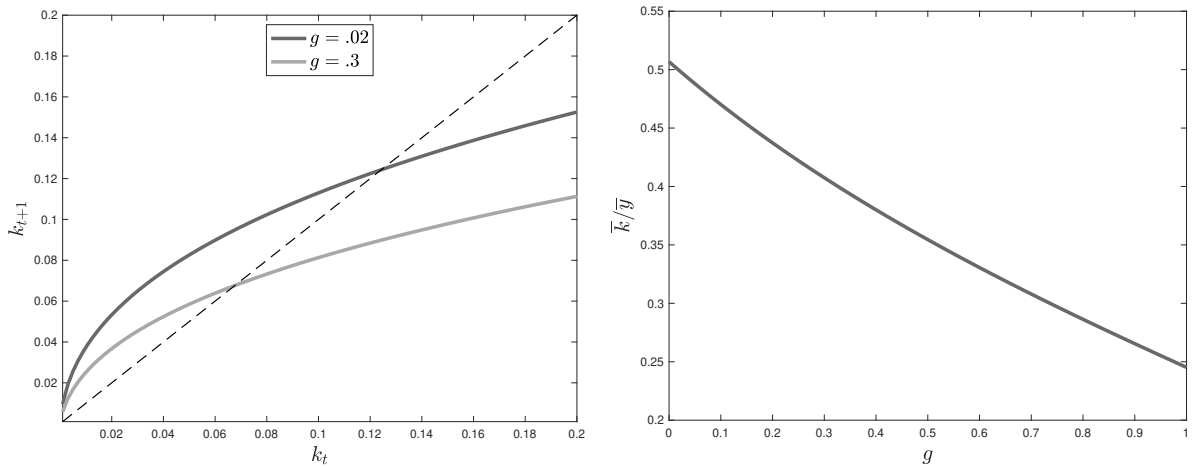
A.3 Putting together the expressions for w_t , r_{t+1} , $C_{1,t}$ and $C_{2,t+1}$, find the equilibrium relation $k_{t+1} = \phi(k_t)$.

- A.4** Plot the relation $k_{t+1} = \phi(k_t)$ in the plane (k_t, k_{t+1}) . Show the steady state $\bar{k}$ and show that it is stable. Compute $\bar{k}$ analytically.
- A.5** Compute $\bar{k}/\bar{y}$ where $\bar{y}$ is steady state output (per efficient unit of labour). How does it vary with g ? What is its limit when $g \rightarrow 0$. Discuss in relation with Piketty.
- A.6** Compute the ratio of consumption of the young of period t over consumption of the old of period t at the steady state. How does consumption inequality across age vary with g ? Explain.
- A.7** Assume now that utility is

$$U = \frac{C_{1,t}^{1-\theta}}{1-\theta} + \frac{1}{1+\rho} \frac{C_{2,t+1}^{1-\theta}}{1-\theta},$$

with $\rho > -1$ and $\theta > 0$. From the utility maximisation problem, find an expression for the saving rate as a function of r_{t+1} . Is it increasing or decreasing in r_{t+1} ? Explain.

- A.8** With these preferences, and for $\alpha = 1/3$, $\theta = .5$ and $\rho = .2$, one can numerically evaluate the recursion $k_{t+1} = \phi(k_t)$ and the value $\bar{k}/\bar{y}$ for different levels of the growth rate g . This is shown on the figure below. Are results qualitatively different from what you have obtained with log utility and no discounting in preferences?



Solution:

PART A - SOLUTION -

A-1. $\pi_r = K_r^\alpha (A_r L_r)^{1-\alpha} - w_r L_r - r_r K_r$

FOC / K_r : $\alpha K_r^{\alpha-1} (A_r L_r)^{1-\alpha} = r_r \Rightarrow \alpha \left(\frac{K_r}{A_r L_r} \right)^{\alpha-1} = r_r \Leftrightarrow r_r = \alpha k_t^{\alpha-1}$ (1)

/ L_r : $(1-\alpha) K_r^\alpha A_r^{1-\alpha} L_r^{-\alpha} = w_r \Rightarrow (1-\alpha) \left(\frac{K_r}{A_r L_r} \right)^\alpha = \frac{w_r}{A_r} \Leftrightarrow w_r = (1-\alpha) k_t^\alpha$ (2)

and $k_t = \frac{Y_r}{A_r L_r} = k_r^\alpha$

A-2 Intertemporal BC is $C_{1t} + \frac{C_{2t+1}}{1+r_{t+1}} = w_r A_r \Rightarrow C_{2t+1} = (1+r_{t+1})(w_r A_r - C_{1t})$

$\Rightarrow M = \log C_{1t} + \log ((1+r_{t+1})(w_r A_r - C_{1t}))$

max M : FOC $\frac{1}{C_{1t}} = \frac{1+r_{t+1}}{C_{2t+1}} \Rightarrow C_{2t+1} = (1+r_{t+1}) C_{1t}$

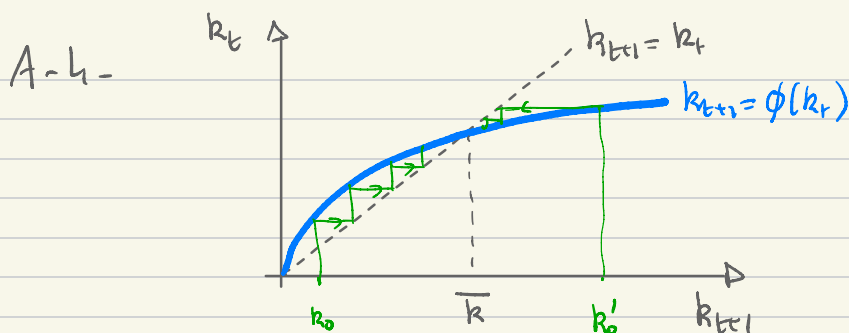
Plug in the IBC to obtain $C_{1t} = \frac{w_r A_r}{2}$ and then $C_{2t+1} = (1+r_{t+1}) \frac{w_r A_r}{2}$

income in period 1 is $w_r A_r$. The saving rate is $1 - \frac{\text{consumption}}{\text{Income}} = \frac{1}{2}$

The saving rate is 50%. It is independent of r_{t+1} , because with log-utility, income and substitution effects offset each other.

A-3 In equilibrium, period $t+1$ capital is period t savings: $K_{t+1} = C_{1t} = \frac{1}{2} w_r A_r$

$\Rightarrow \frac{K_{t+1}}{A_{t+1}} = \frac{1}{2} \overbrace{(1-\alpha) k_t^\alpha}^{w_r} \underbrace{\frac{A_t}{A_{t+1}}}_{\frac{1}{1+g}} \Rightarrow K_{t+1} = \frac{1-\alpha}{2(1+g)} k_t^\alpha = \phi(k_t)$



→ = dynamics ($\bar{k}$ is stable)

Analytically, $\bar{k}$ satisfies $\bar{k} = \phi(\bar{k}) \Rightarrow \bar{k} = \left(\frac{1-\alpha}{z(1+g)} \right)^{\frac{1}{1-\alpha}}$, $\bar{k}$ decreases with g .

A.5 $\bar{y} = \bar{k}^\alpha$ so that $\frac{\bar{k}}{\bar{y}} = \bar{k}^{1-\alpha} = \frac{1-\alpha}{z(1+g)}$

$$\frac{d(\bar{k}/\bar{y})}{dg} < 0 \quad \text{and} \quad \frac{\bar{k}}{\bar{y}}(g \rightarrow 0) = \frac{1-\alpha}{2}$$

$\frac{\bar{k}}{\bar{y}}$ does not go to $+\infty$ when g goes to zero, but decreases towards a constant.

This is different from PIGERY, because here $\frac{\bar{k}}{\bar{y}}$ is derived from a fully specified model.

A.6. Ratio $R = \frac{C_{1t}}{C_{2t}} = \frac{w_t A_t - k_{t+1}}{(1+r_t) k_t} = w_t - \frac{A_{t+1} k_{t+1}}{A_t k_t} \quad \text{and} \quad \bar{k} = \left(\frac{1-\alpha}{z(1+g)} \right)^{\frac{1}{1-\alpha}}$

$$\text{At the ss, } R = \frac{(1-\alpha)k^\alpha - \frac{k}{1+g}}{(1+\alpha k^{1-\alpha})k} = \frac{(1-\alpha)k^\alpha - (1+g)k}{\alpha k^\alpha + k} = \frac{(1-\alpha)k^{1-\alpha} - (1+g)}{\alpha k^{1-\alpha} + 1}$$

$$= \frac{(1-\alpha) \frac{z(1+g)}{1-\alpha} - (1+g)}{\alpha \frac{z(1+g)}{1-\alpha} + 1} = \frac{(1+g)(2-1)}{\frac{1}{1-\alpha}(2\alpha(1+g)+1-\alpha)} = \frac{(1-\alpha)(1+g)}{\alpha + 2\alpha g + 1}$$

$$\frac{\partial R}{\partial g} \approx \frac{(1+g)(2-1) - (1+g)2\alpha}{(\alpha + 2\alpha g + 1)^2} = \frac{1-\alpha}{(\alpha + 2\alpha g + 1)^2} > 0$$

↑ same sign than " Young consume relatively more than old when g is higher.

$$A.7. \quad U = \frac{C_{1t}^{1-\theta}}{1-\theta} + \frac{1}{1+\rho} \frac{1}{1-\theta} \left(\underbrace{(1+r_{t+1})(w_t A_t - C_{1t})}_{C_{2t+1}} \right)^{1-\theta}$$

$$F.O.C.: \quad C_{1t}^{-\theta} - \frac{1+r_{t+1}}{1+\rho} C_{2t+1}^{-\theta} = 0 \quad \Rightarrow \quad C_{2t+1} = \left(\frac{1+r_{t+1}}{1+\rho} \right)^{1/\theta} C_{1t}$$

$$\text{and the IBC gives } C_{1t} + \frac{(1+r_{t+1})^{1/\theta}}{(1+\rho)^{1/\theta}} C_{1t} = A_t w_t \Rightarrow C_{1t} = \frac{(1+\rho)^{1/\theta}}{(1+\rho)^{1/\theta} + (1+r_{t+1})^{1/\theta}} A_t w_t \Rightarrow s(r_{t+1}) = \frac{(1+r_{t+1})^{1/\theta}}{(1+\rho)^{1/\theta} + (1+r_{t+1})^{1/\theta}}$$

$\underbrace{\hspace{10em}}_{1-s(r_{t+1})}$

$$s(r_{t+1}) = 1 - \frac{(1+\rho)^{1/\theta}}{(1+\rho)^{1/\theta} + (1+r_{t+1})^{1/\theta}} \text{ is increasing in } r_{t+1} \text{ if } (1+r_{t+1})^{1/\theta} \text{ is increasing in } r_{t+1} \Rightarrow \boxed{\theta < 1}$$

When θ is < 1 , there is a lot of intertemporal substitution between C_{1t} and $C_{2t+1} \Rightarrow s' > 0$.

A.8. Results are qualitatively the same: $\frac{\bar{K}}{\bar{Y}}$ decreases when $g \nearrow$ and reaches a positive limit.

$\rightarrow$ we do not get the Piketti result.

PART B : Question

IN A RECENT article in the Financial Times, (28/11/2022), Olivier Blanchard (former Professor at MIT and IMF Chief Economist) wrote “ *Back in 2010, Giovanni Dell’Ariccia, Paolo Mauro and I argued for a 4 per cent target for inflation. [...] On the benefit side, a higher target and, by implication, higher average nominal interest rates would give more room for monetary policy to decrease interest rates when needed, reducing the risk of being constrained by the zero lower bound.*”. What is the point of raising the inflation target? Why and how is the zero lower bound a constraint for monetary policy? Write a structured short essay of no more than 500 words (less is fine, equations and graphs are also welcome) and use what we have seen in class in the essay.

Solution:

- Monetary policy aims at price stability, but is also a tool used to smooth business cycles.
- Typically, monetary policy is lenient in bad times, so that the nominal interest rate is kept low, which is likely to favour consumption and/or investment in the short run.
- But when the nominal interest rate is at its lower bound (that is zero (“*zero lower bound*”) in a world without risk but that can be negative (“*effective lower bound*”) in a risky world), there is no room for a further decrease of the nominal interest rate, so that monetary policy is inefficient. This is the so-called *liquidity trap*. The existence of a lower bound for the nominal interest rate is therefore a constraint for monetary policy.
- By raising the target to 4%, the proposal of Blanchard et al. aims at giving more room for manoeuvre to lower the nominal interest rate in recession.
- This was shown in class in two models: IS-LM and the micro-founded model of KRUGMAN (models can be shortly described).
- Monetary policy can regain efficiency if it can manage inflation expectations (often referred to as “*forward guidance*”)(the real interest can be lowered by lowering the nominal interest rate or by increasing inflation expectations), or by conducting some non-conventional actions (quantitative easing). We haven’t formally seen a model of the later.

PART C : Discussion

ON THE NEXT pages is an article published in The Economist in July 2023 and entitled “In Xi Jinping’s China, central planners rule”. As always in The Economist, the article is not signed. Read carefully this article before answering the questions below. For each question, you should propose a structured answer and use the material we have seen in class.

C.1 Identify sentences that illustrates three of the problems that central planning has to face. (no more than 500 words, less is fine, equations and graphs are allowed)

Solution: Here are four possibilities:

- The Planner may not have enough information about agents preferences. They may decide that some goods (rice) are good for the people and some (fruits or flowers) are not: *“To that end, the supreme leader and his underlings emphasise the need to grow grain, rather than frivolities like fruit or flowers, on China’s limited stocks of prime arable land.”*
- The Planner may not properly set prices, so that producers may not be able to meet both ends when following the plan objectives: *“In part thanks to top-down price controls it is hard for farmers to make money growing staples.”*
- The Planner may not have enough knowledge of the technology (here how manage land) when allocating objectives to producers (here farmers): *“In the southern province of Yunnan, terraced rice fields have been cut into slopes.”*
- The planner may not be benevolent, and the plan objective can be to satisfy the planning body rather than the people: *“Last year that sideline took him to a different village to uproot citrus trees to make arable fields. Asked how farmers reacted, he smiled. “It’s the government’s will, how could they not be satisfied?” he replied.”*

C.2 When can Central Planning be a good thing? (no more than 500 words, less is fine, equations and graphs are allowed)

Solution: Central Planning can be a good when market forces don’t work efficiently:

- When there is monopoly power in an industry. In that case, we know that the market solution is not efficient. One possibility is to regulate competition in that industry by making sure that entry is possible and that there is no abuse of dominant position. Another possibility is to nationalize that industry. The later possibility has been chosen at some point in time in many market economies for industries like electricity provision or train transport.
- When there is no market. When there are asymmetries of information, the market may not exists (typical theoretical example is the markets for used cars studied by Akerlov). For example, there is no insurance market to insure against unemployment risk. Why? Because it is difficult to make sure that unemployment is involuntary. Int that case, the government may intervene and create public unemployment insurance.
- When there are externalities, planning may be preferred. Greenhouse gas emissions is a good example. In this example, the allocation of pollution permits or emission quotas centrally allocated can internalize externalities.

There are also situations in which planning can do as well as the market. Typically, in a period of reconstruction after a war, needs are clearly identified and a central agency can efficiently allocate resources to these needs. To various extent, this mix between market and planning has been followed after WWII by the UK, France, Germany of Japan.

C.3 How can one use a model to assess the efficiency of Central Planning? Is the method bullet-proof? (no more than 500 words, less is fine, equations and graphs are allowed)

Solution: We have seen in class the study of Cheremukin et al. (2016) on the Russian experience during the pre-war Stalin industrialisation (1928-1940). The analysis is done as follows:

- Assume preferences and technology.
- Solve for the efficient path.
- Use the data to calibrate the model parameters (production function, preferences).
- Then use the actual times series to measure the distance between what a benevolent planner would have done and what have happened. That distance is measured as a wedge.
- By examining the wedges, one can measure productivity and identify the inefficiencies in the allocation of inputs (capital and labour) across sectors and in the consumption/saving decisions.
- The method relies heavily on the model chosen to measure the wedges. The measure is as good as the model is to describe the reality of a given economy.
- One can then give a brief account of the Cheremukin et al. (2016) paper on Russian industrialization.

“IN XI JINPING’S CHINA, CENTRAL PLANNERS RULE”
When one plan misfires, expect another laid on top
The Economist, Jul 27th 2023



THE PROBLEM with central planners is not that they make mistakes. After all, everyone is fallible: even (oh, the shame of it) newspaper columnists. The trouble with technocrats is how they respond when plans go awry. All too often, when goals are missed or policies backfire, their solution is another plan laid on top.

This dynamic is increasingly visible in the China ruled by Xi Jinping. And it is especially true in the countryside. Policies for rural areas have become a tangled thicket of clashing priorities, core principles and “red lines” that must be defended. The transformative reforms of the 1980s, when collective farms were broken up and peasants were allowed to plant whatever the market was eager to buy, seem ever further away.

This renewed insistence on planning comes from the top. Time and again, and most recently at a meeting on July 20th of the powerful Central Finance and Economics Commission, Mr Xi has called food security a *guozhidazhe*. That is the high-flown phrase which he uses to denote “the main affairs of state” or “national priorities”. To that end, the supreme leader and his underlings emphasise the need to grow grain, rather than frivolities like fruit or flowers, on China’s limited stocks of prime arable land.

Chinese leaders have long worried about feeding nearly a fifth of the people on Earth with 9% of the world’s arable land and 6% of its fresh water. Mr Xi wants China to become less reliant on imports, notably in an age of rising international tensions. As he puts it, “Chinese people should

hold their rice bowls firmly in their own hands, with grains mainly produced by themselves.” In part thanks to top-down price controls it is hard for farmers to make money growing staples. Thus food security challenges another of Mr Xi’s signature policies: identifying high-value crops and agricultural industries to raise farmers’ incomes in the name of poverty alleviation and “rural revitalisation”.

In a growing list of places, officials are responding with more zeal than common sense. Village cadres have sent bulldozers to tear out fruit trees. In the south-western city of Chengdu, Chaguan saw corn and sunflower fields freshly planted in a new suburban park, the Chengdu Eco Belt Park, and along highway verges. In the southern province of Yunnan, terraced rice fields have been cut into slopes. When videos of such incidents surface on social media, some angry netizens recall disastrous Mao-era campaigns to grow food on steep hillsides. Others ask what happened to another national priority since the 1990s, namely the creation of forests to combat soil erosion and bind deserts, including by planting trees on underused farmland beneath the slogan “Grain to Green”.

This resentment is heard in Beijing. In June the Ministry of National Resources issued a directive banning local governments from trying to grow crops on steep slopes and ecologically fragile land, or from using “one-size-fits-all” measures like bulldozing orchards, nurseries and fish ponds to generate new farmland. When recovering arable land that was illegally taken for construction or for other money-earning schemes, officials must proceed “step by step” and protect farmers’ interests at all times.

Yet on the ground a bossier approach may be seen, involving new plans on top of failed schemes. In and around Chengdu, the capital of Sichuan province, officials are focusing on Mr Xi’s current priority. That means growing grain. They are making their obedience visible. Mr Xi visited the fertile Chengdu plain last year, recalling how it was known in history as “heaven’s granary”. Those words, along with Xi-isms about food security and cropland, now appear on village walls and roadside propaganda posters.

The road to the fourth section of Pingshan village, south of Chengdu, dips and climbs like a fairground ride. Every inch of the reddish soil is used. Tiny plots are planted with pungent Sichuan peppercorns, plum trees, grapevines and tea bushes. Still, incomes are low. In 2011 officials boasted of following a priority of the day by “returning farmland to forests”. In Pingshan, this involved planting mulberry trees and promoting the rearing of silkworms. State media predicted that this would open up “new ways to get rich”. Yet today the mulberry trees are gone. Some excited netizens claim that China is in the grip of a national deforestation campaign. The reality is untidier. Mulberry trees did not make money so villagers stopped growing them, reports a young, bare-chested man in sunglasses, watched by a toddler on a tricycle. But Pingshan is not caught up in mass grain-growing campaigns either, being too dry and far from a main road, the young man adds. Luckily, he has another job driving an excavator. Last year that sideline took him to a different village to uproot citrus trees to make arable fields. Asked how farmers reacted, he smiled. “It’s the government’s will, how could they not be satisfied?” he replied.

Because the party knows best

In nearby Yueying village, locals talk of a neighbour who was reluctant to give up his trees. Because the man was a party member, however, he was told “to lead the way” and submit. In Gaohe village, a farmer surnamed Luo describes years of policy about-turns. “When I was young, we grew rice here,” Mr Luo explains. Then locals were encouraged to rent their small plots to a company growing grapes and other cash crops. “That didn’t work out, so they returned the land to us and we started growing citrus fruit,” he says. His family’s six mu (0.4 hectares) of land supported a few hundred trees in all. They made up to 30,000 yuan (\$4,200) per mu in good years. The trees were uprooted last year and the land handed to a company that grows rice in a combined plot of about 20 hectares. Mr Luo’s family receives 1,200 yuan per mu in annual compensation. He is fatalistic about clever schemes that come and go. Farming supplements his main income from working in a nearby town. A few trees still surround his family home. “The country asks us to grow rice, so we grow rice,” he says. In Mr Xi’s China, changeable policies are like the weather: a constant to be endured.